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Bendigo North Soccer Club

Building User Guide

OFFICIAL DIGITAL PORTAL

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Welcome & Building Overview

Achieving energy efficiency is vital for the building. This facility is equipped with several sustainability features, including a 9kW rooftop Solar PV system, efficient lighting and air conditioning, and rainwater harvesting for potable water reductions.

Building users play a critical role in maximizing this efficiency. Please refer to this Building User Guide for practical information on how to stay comfortable while managing the building’s operational efficiency.

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Users are also encouraged to regularly review metering and sub-metering data to better understand energy and water usage patterns. You will find it helps you to better identify opportunities for cost savings, rather than relying solely on utility bills for performance feedback.

Water Consumption & Shower Efficiency

Water Efficient Fittings

New fittings and fixtures installed in this facility are water efficient types.

The star ratings for these fittings are nominated below:

- Basin taps: 6 stars
- Dishwashers (domestic type): 4 stars
- Kitchen taps: 4 stars
- Showers: 4 stars (> 6.0 but <= 7.5 lpm)
- Toilets: 4 stars

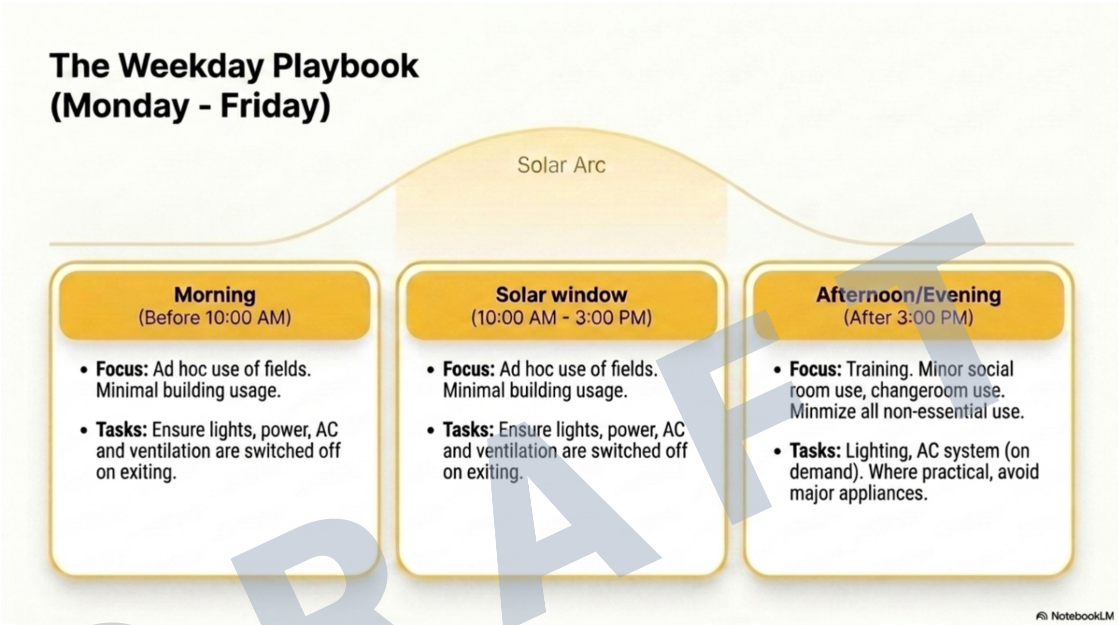
Hot Water and Shower

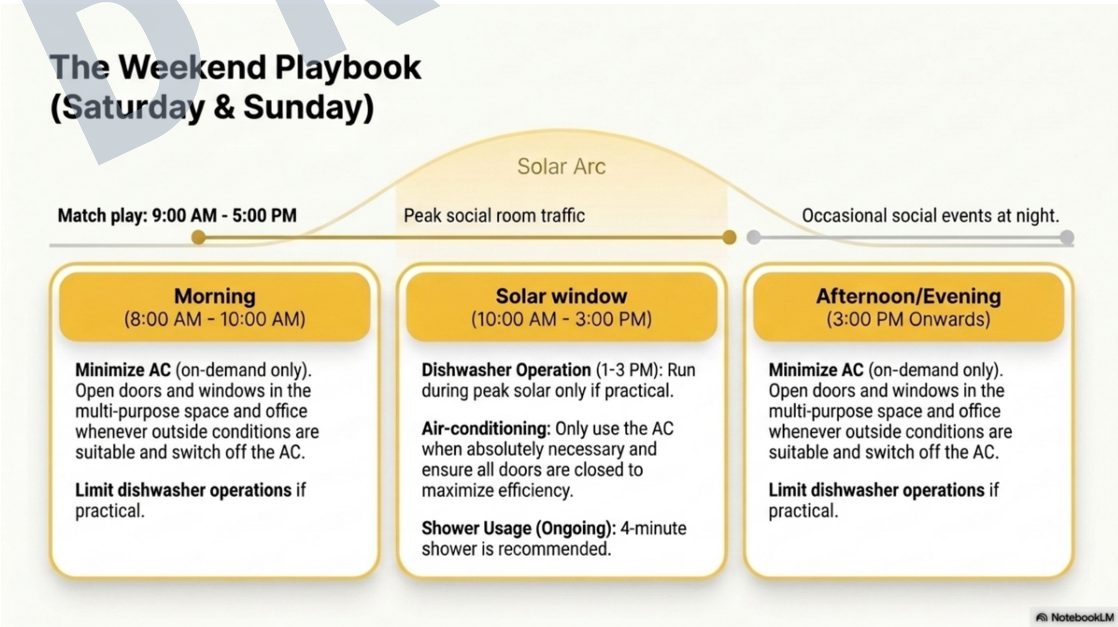
- **Water Temperature**: Water temperature for showers are factory-preset at 45°C maximum. Temperature adjustments must only be made by authorised personnel for legal and safety reasons.
- **Solar Maximize:** Hot water is most efficient when used between 10:00 am and 3:00 pm. This allows the unit to utilise any available "free" rooftop solar power rather than the grid.
- **Preventing Hot Water Shortages:**
    - **Shorter Showers**: Aim for a **4-minute shower**. Reducing shower time from 7 minutes to 4 can save up to 10 litres of hot water per person.
    - **System Recovery:** The hot water storage allows a maximum of 42 showers. Once this is used up, allow system recovery for **1 hour**, which **should be enough for another two teams** showering if a 4-minute shower is adopted.


SHOWERS

Smart Energy Scheduling Guide

This facility operates with a 9kW solar power system. To maximize energy efficiency and reduce costs, where practical, energy-intensive activities should be scheduled during daylight hours when solar power is available (10:00 AM - 3:00 PM). This guide outlines recommended scheduling for common building operations.





Climate Comfort & Efficiency

ON/OFF and Temperature Control

Wall-mounted thermostat controllers are provided in the Multi-purpose Room, Office, and Foyer, allowing the air conditioning system in each area to be controlled independently. Users can operate the air conditioning in each room using the local thermostat controller to turn the unit on or off, adjust the temperature, and control airflow settings.

Temperature Set-Points

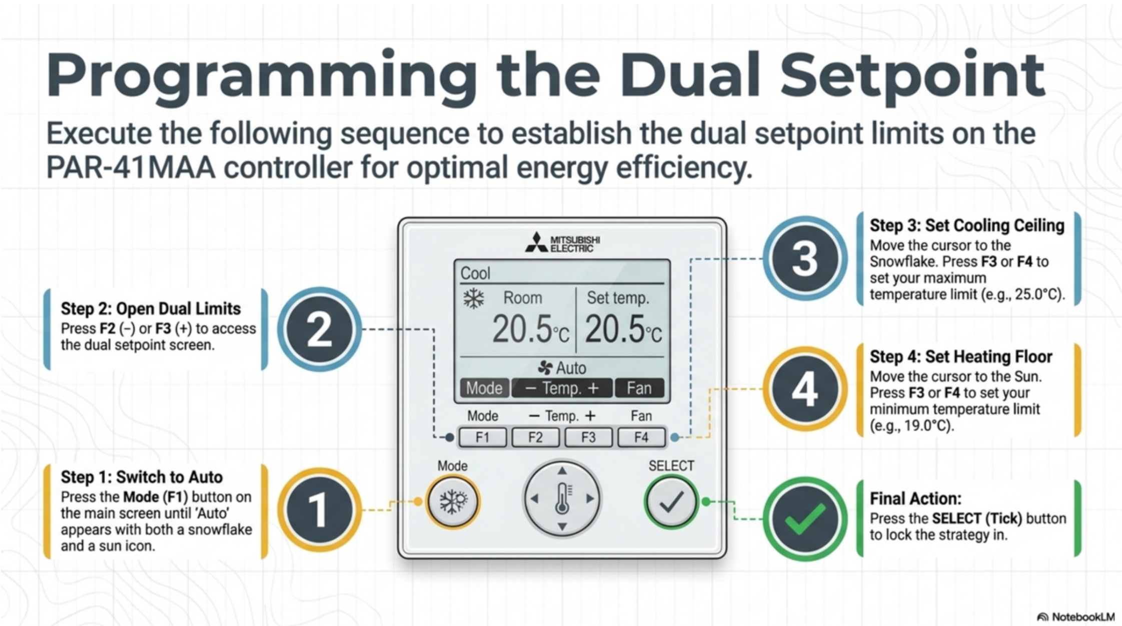
To balance occupant comfort with the building's energy efficiency targets, the system is designed to the following seasonal set-points:

- Summer: 23°C (+/- 2°C)
- Winter: 21°C (+/- 2°C)

Energy saving

However, the system has many energy-saving features that can be learned from the Instruction Book. A handy feature is the Auto (dual set point) mode.

By setting separate heating and cooling targets (e.g., heating 19°C and cooling 25°C), you create a natural "floating zone". The air conditioner stays completely idle as long as the room temperature stays within this buffer, preventing the system from trying to meet a strict temperature range. The lower that you can set the heating temperature and the higher the cooling temperature, the more energy you will save.



For additional guidance on operating the air-conditioning system, consult the PAR-41MAA Instruction Book (searchable on Google using the phrase "PAR-41MAA Instruction Book").

When the HVAC System is ON…

- Keep external doors closed while the system is running during extreme temperatures to maintain thermal efficiency.
- Make sure that both ceiling units in the Multipurpose Room are set to the identical operating mode (Heating or Cooling).
- External openings to conditioned spaces can cause significant energy loss. When air conditioning is operating, manage kiosk windows and doors carefully, avoid propping doors open, and where external patron demand is low, consider serving from inside .

End-of-Day Procedures

Ensure all wall-mounted controllers are manually switched off to maximize building energy savings. This can be performed on the local units or by the central controller in the Bar.

Perform a final manual check to ensure that all exhaust systems are shut off except the essential ones (eg: the ones in the storeroom).

HeadingMulti-Purpose Space Outdoor Air

The Multi-Purpose Space is supplied with outdoor air via the air conditioning units. When the air conditioners in this space operate, a fan for that system also operates to provide outside air directly into the air conditioner air circulation path. This is an automatic feature that cannot be overridden.

Changeroom Heat Recovery Ventilation

To keep your change rooms constantly fresh and dry, a heat recovery ventilation system operates to pull in outdoor air while exhausting stale, humid indoor air.

Its clever feature is a built-in energy recovery core. Before throwing away the comfortable indoor air, the core extracts that temperature and transfers it into the incoming outdoor air stream.

By pre-warming or pre-cooling the incoming fresh air for free, you get maximum ventilation and odor control without wasting energy or making the change rooms drafty.

- There are 2 types of controller for Mitsubishi Sensible Core Lossnay HRVs, see page 9

Exhaust Fans

Ventilation within the changerooms showers, restrooms, and cleaner room is automated via the lighting sensors, with each exhaust fan set to a 10-minute run-down timer.

Specialized units, such as those for the dishwasher and storeroom, are operated by localized manual switches clearly marked with their respective labels.

Waste Management

Internal Lighting

Internal Lighting

The lighting system is designed to improve energy efficiency by responding automatically to occupancy and available natural daylight.

All rooms and internal spaces include occupancy and daylight sensors, which reduce unnecessary lighting use by switching lights off when spaces are unoccupied and adjusting lighting levels when sufficient natural daylight is available.

Additional controls include:
- Lighting throughout the building is controlled through a central switching panel. A master switch labelled “Master” allows staff to switch off general lighting when the building is not in use, helping reduce unnecessary energy use.
- An after-hours switch is also provided, allowing selected lights to remain on temporarily for up to two hours outside normal operating times.
- Bar/Kiosk lighting: controlled by a time clock with manual switching capability.

External Lighting

External lighting is controlled by a combination of a photoelectric cell switch and a time clock. The photoelectric cell, mounted at high level on the south-facing wall, automatically switches the lighting on and off based on ambient daylight levels. The time clock provides additional scheduling control and can be set to Auto, Manual, or Changeover with Off settings, allowing flexibility for automatic operation or manual override as required.
- External circulation/access lighting: controlled by a time clock and PE daylight sensor.
- External security lighting: controlled by a PE daylight sensor.

Access Control & Security

The building security system is a Windows-based access control system with real-time monitoring and site plan integration. Operators can view device locations and status, remote release front door, and manage proximity cards through the security panel within the Communication Room. Detectors can be armed/disarmed from the keypad within entry.

All main entry points are fitted with electronic strike locks and integrated with the building's central security system.

- Staff Access: Use your assigned encrypted key fob at the proximity readers located at the North and South entries.
- Visitor Access: During business hours, visitors should use the intercom located at the North Gate to speak with reception.